

Po-Chen Ko

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Education

National Taiwan University

Bachelor of Science in Engineering, Electrical Engineering (EE)

GPA: 4.08/4.3

Relevant Courses: Machine Learning, Computer Vision, Applied Deep Learning.

Taipei, Taiwan

Sep. 2020 - Aug. 2024

Research

Learning To Act From Actionless Videos Through Dense Correspondences

Po-Chen Ko, Jiayuan Mao, Yilun Du, Shao-Hua Sun, Joshua B. Tenenbaum

ICLR 2024 (**Spotlight**)

📄 [paper](#) 🌐 [project page](#) 📄 [Code](#)

Context-Aware Replanning with Pre-Explored Semantic Map for Object Navigation

Po-Chen Ko*, Hung-Ting Su*, Ching-Yuan Chen*, Jia-Fong Yeh, Min Sun, Winston H. Hsu

CoRL 2024

📄 [paper](#) 🌐 [project page](#) 📄 [Code](#)

Implicit State Estimation via Video Replanning

Po-Chen Ko, Jiayuan Mao, Yu-Hsiang Fu, Hsien-Jeng Yeh, Chu-Rong Chen, Wei-Chiu Ma, Yilun Du, Shao-Hua Sun

ICML Workshop on Building Physically Plausible World Models 2025 (**Best paper**), Under Submission

📄 [paper](#) 🌐 [project page](#) 📄 [Code](#)

Learning Skills from Action-free Videos

Hung-Chieh Fang*, Kuo-Han Hung*, Chu-Rong Chen, Po-Jung Chou, Chun-Kai Yang, Po-Chen Ko, et al.

ICML Workshop on Building Physically Plausible World Models 2025, Under Submission

📄 [paper](#)

Experience & Internship

Computer Science and Artificial Intelligence Laboratory (CSAIL), MIT

Cambridge, MA

Visiting Student @ cocosci group advised by Prof. Joshua B. Tenenbaum

Feb. 2023 - Aug. 2023

Led the Development of a framework that trains an RGB observation conditioned diffusion model as a planner and recovers robot actions from predicted videos through dense correspondences.

Robot Learning Lab (RLL), NTU

Taipei, Taiwan

Undergraduate Researcher advised by Prof. Shao-Hua Sun

Feb. 2022 - Present

Gained knowledge about RL and programmatic agents. Explored controlling the behavior of multi-task agents by exploiting multi-dimensional reward function.

Communication and Multimedia Lab (CMLab), NTU

Taipei, Taiwan

Undergraduate Researcher @ MiRA group advised by Prof. Winston Hsu

Feb. 2024 - Apr. 2025

Improved VLM-based semantic maps for Object Navigation by leveraging uncertainty information from the map to aid candidate selection.

Awards & Achievements

Travel Grant Recipient — CoRL 2024	Oct. 2024
1st Place / Department-wide — NTUEE Undergraduate Innovation Award <i>Departmental prize for undergraduate research project</i>	Jun. 2024
200+ GitHub stars — AVDC open-source video-planner training library	Oct. 2023
1st Place / 1170 participants — ML2021Spring Speaker Classification Competition	Jun. 2021

Projects

Coordinate-ResNet: Inductive Bias for Vision Backbones.  Code	Dec. 2023
• Integrated a coordinate-based inductive bias into a ResNet backbone, enhancing spatial awareness in vision tasks. • Evaluated on spatially-rich target tasks and demonstrated significantly improved performance through learned coordinate features and architecture augmentation.	
User Preference Prediction System.  slides	Jan. 2023
• Built a preference-prediction pipeline using augmented user data, text embeddings (mT5, text2vec), and multi-head attention, exploring seq2seq and top-K classification models. • Improved unseen-course prediction through course-similarity modeling (CoSENT) and k-NN score adjustment, yielding consistent gains on validation and test sets.	
Hexapod Trajectory Planning with Actor-Critic Learning.  slides	Oct. 2022
• Designed a trajectory-planning system for a hexapod robot using an Actor-Critic framework, where the critic learned terrain-aware value estimates from simulation data and the actor exploited them to refine motion trajectories. • Improved gait stability and path efficiency by integrating learned value functions with classical trajectory heuristics, achieving smoother and more reliable locomotion in simulation.	
Pupil Tracking with Semantic Segmentation and Confidence Estimation.  report	Jun. 2022
• Designed a lightweight autoencoder with 5-frame temporal input for robust pupil segmentation. • Built a rule-based confidence estimator using pupil-area dynamics and temporal smoothing.	
Othello: A chatbot that uses RL-based policies to play the othello game with user.  Code	Aug. 2021
• Built a modular Othello engine with self-play, move generation, and evaluation tools. • Implemented multiple agents including Monte-Carlo rollouts and a DQN learner, and benchmarked them through scripted tournaments.	